Dynamic Stochastic Modeling
Lecture 4: State-Contingent Claims and Asset Pricing

1 From Uncertainty to Markets

In the previous lecture, we studied how an individual evaluates uncertain consumption. If tomorrow has states $s = 1, \ldots, S$ with probabilities

$$\pi_s > 0, \qquad \sum_{s=1}^{S} \pi_s = 1,$$

then time-separable expected utility can be written

$$U = u(c_0) + \beta \sum_{s=1}^{S} \pi_s u(c_1(s)).$$

Risk aversion gives people a reason to move consumption across states. If consumption is high in one state and low in another, a risk-averse consumer would generally like to give up consumption in the good state in exchange for additional consumption in the bad state.

This raises the central question of this lecture:

> Can markets allow people to trade consumption across states of the world?

The answer leads directly to asset pricing.

2 A Two-State Economy

There are two dates, date 0 and date 1. At date 1, one of two states occurs:

$$H \qquad \text{or} \qquad L,$$

with

$$\pi_H + \pi_L = 1.$$

There are now three distinct commodities:

$$c_0, \qquad c_1(H), \qquad c_1(L).$$

The last two goods are physically identical and delivered at the same date. They are nevertheless different economic commodities because they are delivered under different circumstances.

Thus a commodity can be defined not only by what it is and when it is delivered, but also by the state of the world in which it is delivered.

3 State-Contingent Claims

Suppose that at date 0 we can purchase a claim that pays one unit of consumption at date 1 if state H occurs and zero otherwise. Its payoff is

$$(1, 0).$$

Similarly, a claim with payoff

$$(0, 1)$$

delivers one unit in state L and nothing in state H.

These are called *Arrow securities* or *Arrow–Debreu securities*.

Let q_H be the date-0 price of one unit of consumption delivered in state H, and let q_L be the date-0 price of one unit delivered in state L. These are *state prices*.

The price q_L is not simply the price of consumption tomorrow. It is the price today of consumption tomorrow *conditional on state L occurring*.

4 The Consumer's Problem

Suppose the consumer has endowment

$$(e_0, e_1(H), e_1(L)).$$

The *expected* present value of the endowment is

$$e_0 + q_H e_1(H) + q_L e_1(L).$$

The budget constraint is therefore

$$c_0 + q_H c_1(H) + q_L c_1(L) \leq e_0 + q_H e_1(H) + q_L e_1(L).$$

This expression uses date-0 prices.

The consumer solves

$$\max_{c_0, c_1(H), c_1(L)} \{u(c_0) + \beta [\pi_H u(c_1(H)) + \pi_L u(c_1(L))]\}$$

subject to this budget constraint.

The Lagrangian is

$$\begin{aligned}\mathcal{L} = {} & u(c_0) + \beta [\pi_H u(c_1(H)) + \pi_L u(c_1(L))] \\ & + \lambda [e_0 + q_H e_1(H) + q_L e_1(L) - c_0 - q_H c_1(H) - q_L c_1(L)].\end{aligned}$$

For an interior solution,

$$u'(c_0) = \lambda,$$

$$\beta \pi_H u'(c_1(H)) = \lambda q_H,$$

and

$$\beta \pi_L u'(c_1(L)) = \lambda q_L.$$

Therefore

$$\boxed{q_H = \beta \pi_H \frac{u'(c_1(H))}{u'(c_0)}}$$

and

$$\boxed{q_L = \beta \pi_L \frac{u'(c_1(L))}{u'(c_0)}.}$$

More generally,

$$\boxed{q_s = \beta \pi_s \frac{u'(c_1(s))}{u'(c_0)}.}$$

This is our first fundamental asset-pricing equation.

5 Probability Is Not the Same as Price

The probability π_s tells us how likely state s is to occur. The state price q_s tells us how much the consumer is willing to pay today for one unit of consumption delivered in that state.

The first-order condition gives

$$\frac{q_s}{\pi_s} = \beta \frac{u'(c_1(s))}{u'(c_0)}.$$

Thus equally likely states need not have equal state prices.

Suppose

$$\pi_H = \pi_L$$

but

$$c_1(L) < c_1(H).$$

For a risk-averse consumer,

$$u'(c_1(L)) > u'(c_1(H)),$$

and hence

$$q_L > q_H.$$

A unit of consumption delivered in the low-consumption state is more valuable than a unit delivered in the high-consumption state.

The important lesson is:

> An asset is valuable not merely because it pays a lot, but because of when and under what circumstances it pays.

6 From State Claims to Assets

Suppose an asset pays x_H units of consumption in state H and x_L units in state L. Its payoff vector is

$$x = (x_H, x_L).$$

If state claims can be traded separately, this asset is a package containing x_H units of the H claim and x_L units of the L claim. Its date-0 price must therefore be

$$\boxed{p = q_H x_H + q_L x_L.}$$

With S states,

$$\boxed{p = \sum_{s=1}^{S} q_s x_s.}$$

The price of an asset is the state-price-weighted value of its payoffs.

7 The Stochastic Discount Factor

Substituting the state-price formula into the asset-price formula gives

$$p = \sum_s \pi_s \left[\beta \frac{u'(c_1(s))}{u'(c_0)} \right] x_s.$$

Define

$$\boxed{m_s = \beta \frac{u'(c_1(s))}{u'(c_0)}.}$$

The random variable m is called the *stochastic discount factor*, or the *pricing kernel*. Then

$$\boxed{p = E[mx].}$$

The stochastic discount factor is high in states in which marginal utility is high. For a risk-averse consumer, those tend to be states in which consumption is low. Payoffs received in low-consumption states therefore receive large weights in asset prices.

8 A Risk-Free Asset

Consider an asset that pays one unit of consumption tomorrow in every state. Its payoff vector is

$$(1, 1, \ldots, 1).$$

If its date-0 price is p_f, its gross risk-free return is

$$R_f = \frac{1}{p_f}.$$

But this asset is also a package containing one unit of every state claim. Therefore

$$p_f = \sum_{s=1}^{S} q_s.$$

Hence

$$\boxed{\frac{1}{R_f} = \sum_{s=1}^{S} q_s.}$$

Since

$$q_s = \pi_s m_s,$$

we also have

$$\boxed{\frac{1}{R_f} = E[m]}$$

and therefore

$$\boxed{R_f = \frac{1}{E[m]}.}$$

This connects the uncertain economy directly to the deterministic two-period economy studied in Week 1. There we had one price for future consumption, $1/R$. Here the price of one unit delivered regardless of the state is the sum of the individual state prices.

9 Returns

Suppose an asset has price p today and payoff x_s in state s tomorrow. Its gross return in state s is

$$R_s = \frac{x_s}{p}.$$

Since

$$p = E[mx],$$

division by p gives

$$\boxed{1 = E[mR].}$$

For the risk-free asset,

$$1 = R_f E[m],$$

which again implies

$$R_f = \frac{1}{E[m]}.$$

10 Risk and Expected Returns

Using

$$E[mR] = E[m]E[R] + \mathrm{Cov}(m, R),$$

the equation $1 = E[mR]$ implies

$$1 = \frac{E[R]}{R_f} + \mathrm{Cov}(m, R).$$

Thus

$$\boxed{E[R] - R_f = -R_f \,\mathrm{Cov}(m, R).}$$

An asset whose return is high when m is high pays well when marginal utility is high—typically when consumption is low. Such an asset provides insurance and can have a relatively low expected return.

An asset whose return is low when m is high performs badly precisely when consumption is scarce. Consumers dislike this risk and require a higher expected return to hold the asset.

Thus the relevant question is not simply how volatile an asset is. The more important question is whether the asset pays when consumption is especially valuable.

11 Complete Markets

Suppose there are S possible states tomorrow.

If consumers can buy and sell a separate Arrow security for every state, then they can construct any desired vector of state-contingent payoffs

$$(x_1, \ldots, x_S).$$

Markets are then said to be *complete*.

Complete markets allow consumers, subject to their budget constraints, to move wealth across states of the world. A consumer with a large endowment in one state and a small endowment in another can sell claims to consumption in the high-endowment state and buy claims to consumption in the low-endowment state.

12 Incomplete Markets

If the available assets do not permit consumers to construct every desired state-contingent payoff, markets are *incomplete*.

For example, suppose tomorrow has three states but the only available asset is a risk-free bond. The bond transfers consumption across dates, but its payoff is the same in every state. It cannot transfer consumption specifically from a good state to a bad state.

Thus borrowing and lending can be possible even when complete insurance is not.

This distinction will become important later in the course. In incomplete markets, individuals may face risks they cannot fully insure. Their saving and portfolio decisions then depend on the risks they face as well as on average rates of return.

13 The Main Ideas

The consumer values one unit of consumption in state s according to

$$\boxed{q_s = \beta \pi_s \frac{u'(c_1(s))}{u'(c_0)}.}$$

A state price therefore reflects both probability and marginal utility.

An asset is a collection of state-contingent payoffs:

$$\boxed{p = \sum_s q_s x_s.}$$

Defining

$$m_s = \beta \frac{u'(c_1(s))}{u'(c_0)}$$

gives

$$\boxed{p = E[mx].}$$

Equivalently, in terms of returns,

$$\boxed{1 = E[mR].}$$

Finally,

$$\boxed{\frac{1}{R_f} = E[m].}$$

The central economic message is:

> The value of a future payoff depends not only on how large it is or how likely it is, but also on how valuable consumption will be when the payoff arrives.

This idea will provide the foundation for our later study of asset prices in dynamic economies.